1. Cooling Sleeve
    - Can be used for insulin patches (arm) can also be adapted for standard pumps as well (waist wrapping)
    - Slot for insulin pump to fit into, protected from heat by sleeve materials
    - Cooling gel inside of sleeve, like FRIO wallet
    - Different gel formations --> hexagonal formation for better flexibly/contour to body, gel paste for thicker layering, but could be more unwieldy, more possible gel configurations
    - Needs exposure to water to correctly function, could possibly need continued re-exposure to stay efficient
    - Possible compression sleeve design? --> helps blood flow, leading to more efficient insulin dispersion in the body

2. Cooling gel ring
    - Mainly for insulin patches, not sure how to apply to pumps
    - Cooling ring that fits around the site of the patch, absorbs excess heat that the patch is exposed to
    - Skin side is insulated, outer side contains heat absorbing gel
    - Halo shape around the patch
    - Probably disposable, possible waste issues

3. Mechanical insulin pouch
    - Possible pouch/case with mini fans to funnel out heat away from the pump
    - Potentially bulky, will need electricity/batteries to work

4. Insulin pouch belt
    - Already exists for sleeping nighttime use --> adapt for outdoor workers?
    - Use durable materials, built in insulin case
    - Functions similarly to a phone holster OR the entire belt can have cooling mechanisms woven into it for greater efficiency

5. UV/IR light blockers

- UV is the main carrier of heat from the sun, and IR light is the main source of damage from sunlight (skin cancer, sunburn, etc.)
- Blocking these types of lights could possibly shield the pump/patch from overheating and protect insulin from degrading.
- Pumps are usually made of plastic or polymers, so they are very susceptible to IR light and can result in mechanical failure
- Adhesion is also effected, can be dried out fast and pump/patch can fall off prematurely
- Possible blocker could be a reflective sticker that goes over the pump and blocks harmful light from the sun
- Disposable, must be replaced once a day per use

Isra

1. Smart Insulin Cooling Case

- Portable case that actively regulates insulin temperature before use
- Cooling system built into the case instead of directly on the body
- Bluetooth temperature monitoring with phone alerts
- Maintains insulin within the safe storage range during travel or outdoor exposure
- Prevents insulin degradation before it reaches the pump

Could be something like this: https://inuteq.com/insulin-cooling-bag/?srsltid=AfmBOorOmSdN9O3ZeaADne1T1-3DhgqFk_L2gjomAmfNAH6IOFHpK7aI#blue

2. Bluetooth-Connected Smart Cooling Device

- Wearable cooling module connected to the insulin pump via Bluetooth
- Built-in temperature sensors monitor insulin cartridge heat levels
- Micro cooling chip activates only when temperature exceeds a safe range
- Sends real-time alerts and temperature data to a mobile app
- Power-efficient design that optimizes cooling only when necessary

3. Thermoelectric Cooling Ring

- Ring-shaped thermoelectric device placed around the insulin pump cartridge
- Uses a Peltier cooling element to regulate insulin temperature
- Targets cooling directly at the insulin reservoir location
- Lightweight attachment compatible with tubed insulin pump housings

Prevents insulin overheating in high-temperature environments

4. Heat Shield Patch for Insulin Pumps

- Lightweight patch designed to protect insulin pumps from external heat
- Reflective metallic outer layer reduces solar radiation absorption
- Breathable adhesive layer keeps device comfortable on skin
- Foam insulation layer reduces heat transfer from body temperature

Optional temperature sensor alerts user if insulin becomes too hot

5. Layered Thermal Heat-Redirecting Patch

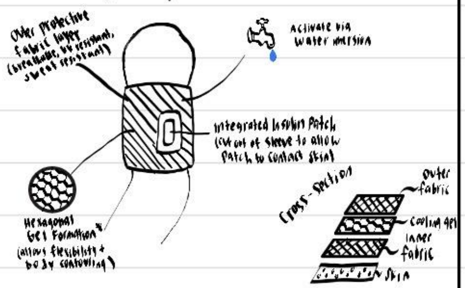
- Multi-layer patch that redirects heat away from the insulin reservoir
- Hydrogel skin interface reduces direct body heat transfer
- Heat spreader network distributes heat to prevent localized hot spots
- Micro phase-change materials absorb sudden temperature spikes
- Radiative outer layer releases heat into the environment

Proposed Product Design Sketches

Mackenzie:

Proposed Product Design Sketches

Cooling Sleeve:



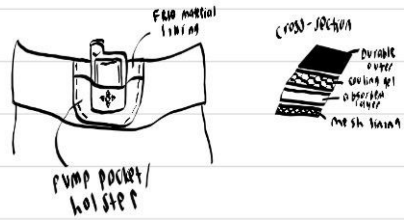
Material ideas for outer layer:
moisture-wicking mesh uses specialized polyester or nylon 3D warp-knitted structures to enhance airflow, manage heat, and pull moisture away from the skin.

Polyester mesh fabrics
- breathable
- durable
- lightweight
- moisture-wicking

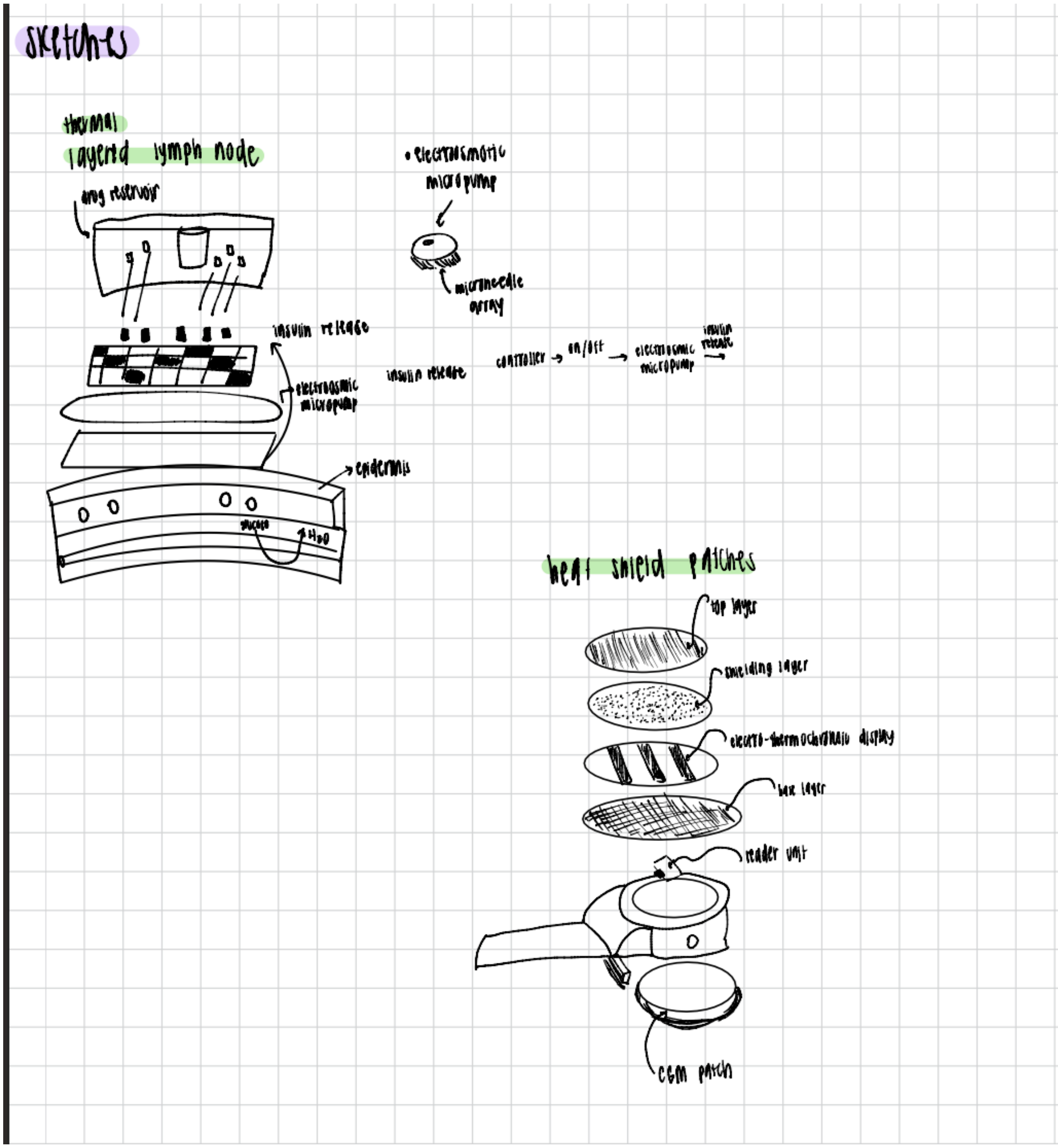
Knowledge Gap: what gel can we specifically use for our project?

* Frio gel is patented

Insulin pouch belt:

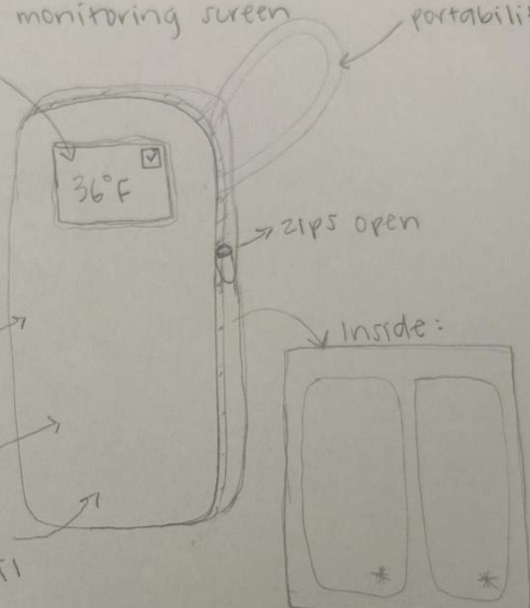


☆ Unsure of what exactly was meant by this proposed design.


sketches
thermal
layered lymph node
drug reservoir
insulin release
electroosmotic micropump
epidermis
glucose
H_2O
• electroosmotic micropump
microneedle array
insulin release
controller → on/off → electroosmotic micropump → insulin release
heat shield patches
top layer
shielding layer
electro-thermochromic display
base layer
reader unit
CGM patch

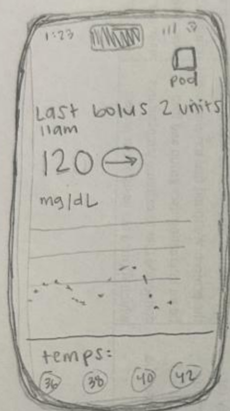
Mary:

Insulin Cooling Case



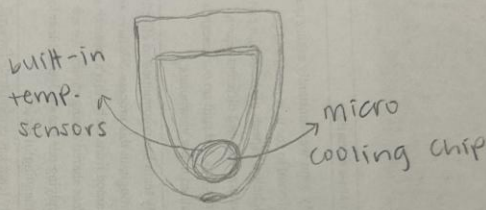
- maintains insulin within a safe storage range (36°F - 46°F) during travel or outdoor exposure
- prevents insulin degradation before it reaches the pump

Bluetooth-Connected Smart Cooling Device



Bluetooth to iPhone (App)

- sends real-time alerts and temp data



- power-efficient design that optimizes cooling only when necessary
- monitors insulin cartridge heat levels

Cooling Sleeve Practical Ideas:

Sensing Layer:

- Use sensor to gather information about conditions affecting insulin stability such as the device temperature or temperature of the outside environment

Processing Layer (Microcontroller):

- The microcontroller will read the data from the sensors and compare values to safe temperature thresholds. From there, it will decide whether cooling needs to increase, decrease, or stop. Coding will allow for this to work.
  Ex: If the temperature exceeds a safe threshold, activate cooling. If the temperature returns to a safe range, reduce cooling.

Actuation Layer (Biggest Concern, need Isra's information on possible gels):

- The microcontroller controls a moisture delivery mechanism that activates or rehydrates the gel. Following the processing layer, actuation will occur by releasing water onto the gel layer to cool the gel.

Feedback/User Interface:

- LED indicator for cooling status
- Warning if insulin temperature becomes unsafe
- Mobile app data display?

UV/IR Light Blocking Sticker Practical Ideas:

Adhesive Layer:

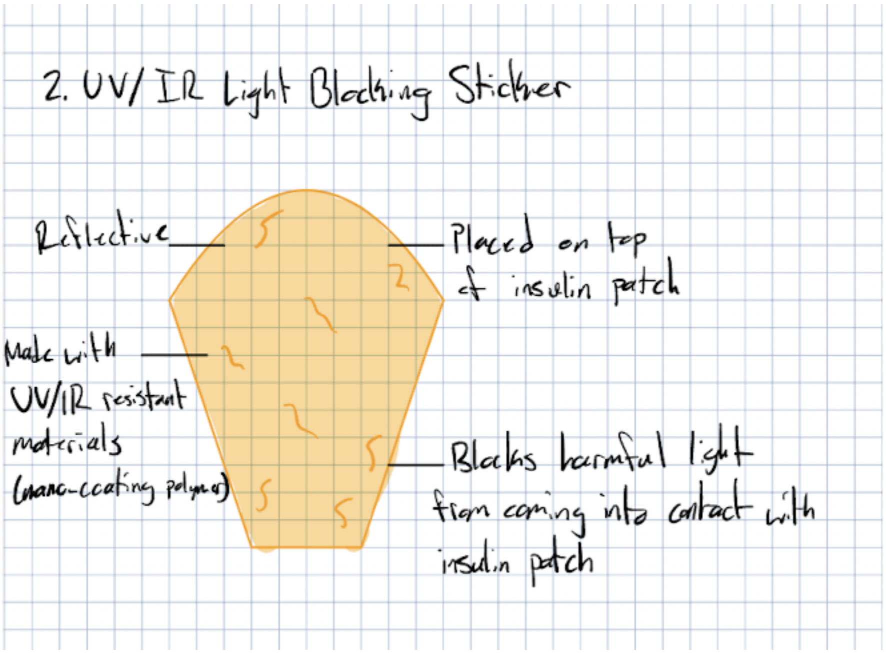
- Designed to stick only to the insulin patch, is not designed to come into contact with the skin
- Must be long lasting and able to stick for multiple hours (bare minimum 24+)

UV/IR Blocking Layer:

- Reflects specifically UV and Infrared light
- Possible materials include aluminum, Metalized Polyethylene Terephthalate, Blackened foils, etc.
- Possible color change when absorbing/reflecting layer is running low on efficiency?

Feedback/User Interface:

- Not sure how to include, perhaps a microchip in each that connects to device to show when sticker runs out of effectiveness?



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